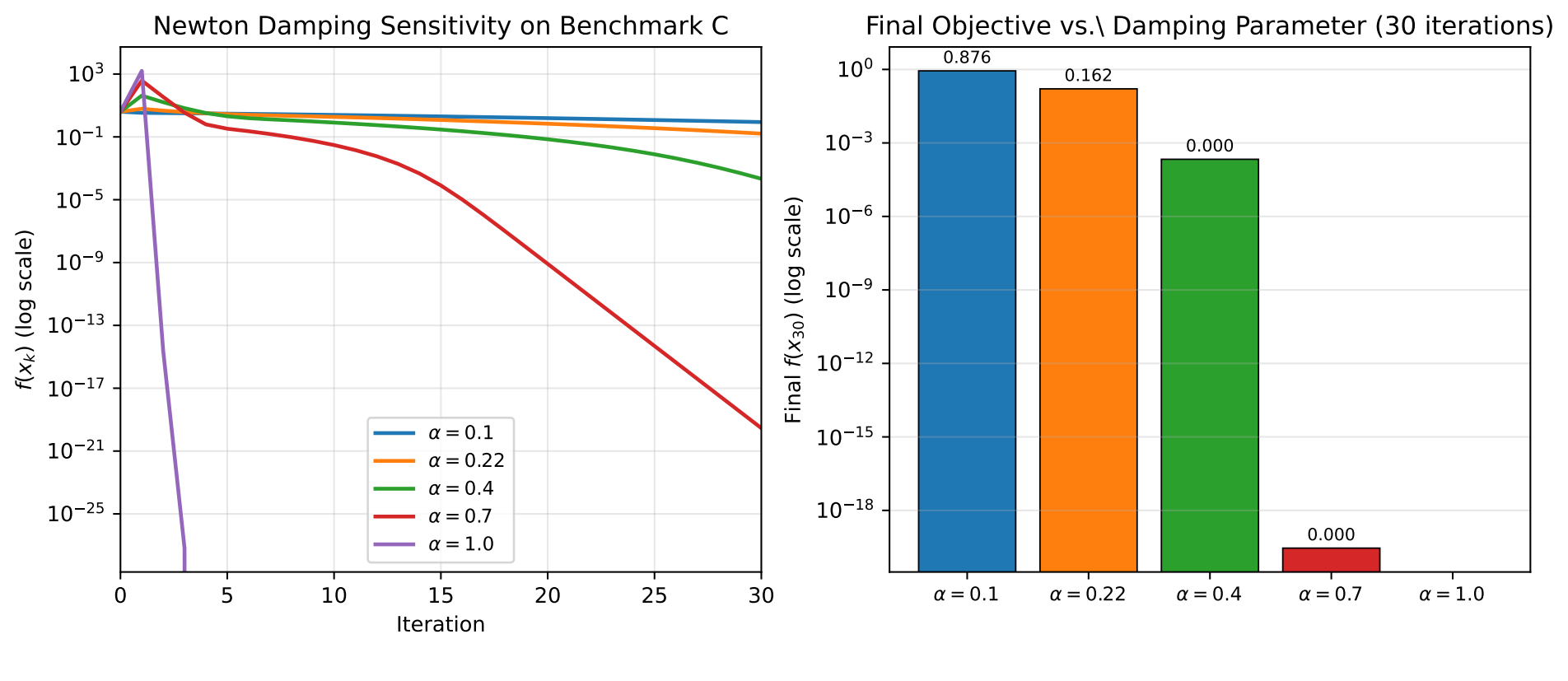
Newton Damping Sensitivity on Benchmark C
$f(x_k)$ (log scale)
10^{3}
10^{-1}
10^{-5}
10^{-9}
10^{-13}
10^{-17}
10^{-21}
10^{-25}
0
5
10
15
20
25
30
Iteration
$\alpha = 0.1$
$\alpha = 0.22$
$\alpha = 0.4$
$\alpha = 0.7$
$\alpha = 1.0$
Final Objective vs.\ Damping Parameter (30 iterations)
Final $f(x_{30})$ (log scale)
10^{0}
10^{-3}
10^{-6}
10^{-9}
10^{-12}
10^{-15}
10^{-18}
0.876
0.162
0.000
0.000
$\alpha = 0.1$
$\alpha = 0.22$
$\alpha = 0.4$
$\alpha = 0.7$
$\alpha = 1.0$
0.000